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# Data Mining

WS 2025/26

## Exercise Sheet 3: Deep Learning on Sets and Graphs

Submission until January 7, 2026, 23:59

Discussion: January 8, 2026

**Exercise 3-1** *Invariant functions and permutation groups* (3 points)

A function $f\colon \mathbb{R}^n \to \mathcal{Y}$ is said to be invariant to a subgroup $G$ of the symmetric group $S_n$ if for all $\pi$ in $G$

$$f(\pi(\boldsymbol{x})) = f(\boldsymbol{x}) \quad \text{for all} \quad \boldsymbol{x} \in \mathbb{R}^n, \tag{1}$$

where $\pi(x_1, x_2, \ldots, x_n) = (x_{\pi(1)}, x_{\pi(2)}, \ldots, x_{\pi(n)})$. A set of permutations $P$ together with the composition of functions denoted by $\circ$ is a *permutation group* if

(i) $P$ is closed under composition, i.e., for all $\pi$, $\sigma$ in $P$ the permutation $\pi \circ \sigma$ is in $P$,
(ii) the identity permutation $i \mapsto i$ for $i$ in $\{1, 2, \ldots, n\}$ is in $P$, and
(iii) for all $\pi$ in $P$ the inverse $\pi^{-1}$ is in $P$.

Let $P \subseteq S_n$ be a set of permutations and $f\colon \mathbb{R}^n \to \mathcal{Y}$ be a function that satisfies Eq. (1) for all $\pi$ in $P$. Show that there is a set $G$ with $P \subseteq G \subseteq S_n$ such that $G$ forms a permutation group and $f$ is invariant to $G$.

**Exercise 3-2** *Implementation of set functions* (1+1+2+2+1 points)

Consider the ground truth function $g\colon \{0, 1, \ldots, 9\}^{10} \to \{-1, +1\}$ with

$$g(\boldsymbol{x}) = \begin{cases} +1 & \text{if there are distinct } i, j, k \in \{1, \ldots, 10\} \text{ with } x_i + x_j + x_k = 5, \\ -1 & \text{otherwise.} \end{cases}$$

We would like to learn the function from training data using a neural network.

a) Write code to create a dataset with class labels according to the function $g$. The ten components of each data point should be drawn uniformly at random from $\{0, 1, \ldots, 9\}$. The number of data points can be adjusted for the later tasks.

b) Compute statistics of your dataset (e.g., class balance) and split the data into training and test sets.

c) Implement a permutation-sensitive neural network and train and evaluate it using your data set. Try different parameters of your network (e.g., learning rate, hidden dimension, number of layers, activation function) and report your results.

d) Implement a permutation-invariant neural network discussed in the lecture (e.g., Deep Sets, Janossy pooling, Deep Sets with attention) and train and evaluate it using your data set. Try different parameters of your network (e.g., realizations of the functions $\rho$ and $\phi$) and report your results.

e) Discuss whether your results in tasks c) and d) meet your expectations. Give reasons for your answer.

**Exercise 3-3** *Neural networks on pairs of sets* (1+1+2+2 points)

Let $\mathcal{U} = \{0, 1, \ldots, 9\}$ and let $\mathcal{X}$ denote the power set of $\mathcal{U}$. Consider the ground truth function $g \colon \mathcal{X} \times \mathcal{X} \to \{-1, +1\}$ with

$$g(A, B) = \begin{cases} +1 & \text{if } A \subseteq B, \\ -1 & \text{otherwise.} \end{cases}$$

We would like to learn the function from training data using a neural network.

a) Write code to create a dataset with pairs of sets and class labels according to the function $g$. You may create a random set by randomly choosing its cardinality $k$ first and then adding $k$ elements selected uniformly at random from $\{0, 1, \ldots, 9\}$. The second set can be generated using the same approach or by modifying the first set. The number of data points can be adjusted for the later tasks.

b) Compute statistics of your dataset (e.g., class balance, distribution of set cardinality). Make sure that you dataset contains a reasonable amount of positive and negative examples. Split the data into training and test sets.

c) Your neural network should compute embeddings for the two sets using a permutation-invariant architecture $f_\theta$. The two embeddings should then be used to obtain a prediction. One method to achieve this is to use a neural network $\text{MLP}(f_\theta(A)^\top f_\theta(B), |A|)$. Implement this approach and train and evaluate it using your dataset. Try different parameters of your network and report your results.

d) Think of other methods to learn functions on pairs of sets. Describe your ideas, implement and compare them to the baseline of task c).